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AWS-DEVOPS



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Introduction to Cloud Computing

- Cloud computing is a technology that allows us to store and process our data and applications in virtual spaces, accessible via the internet, rather than on our physical devices.
- Cloud computing refers to the transition of computer systems from traditional physical hardware to internet-based services. With this technology, data storage, processing power, and software have now become resources that can be accessed remotely via the internet.



What is Cloud Computing?

- Cloud computing is the delivery of IT resources over the internet, on-demand and priced on a pay-as-you-go basis. Instead of purchasing physical data centers and servers, owning them, and ensuring their continuity, you can access computing power, storage, and databases provided by a cloud service provider, such as Amazon Web Services (AWS), whenever you need them.





Who uses cloud computing?

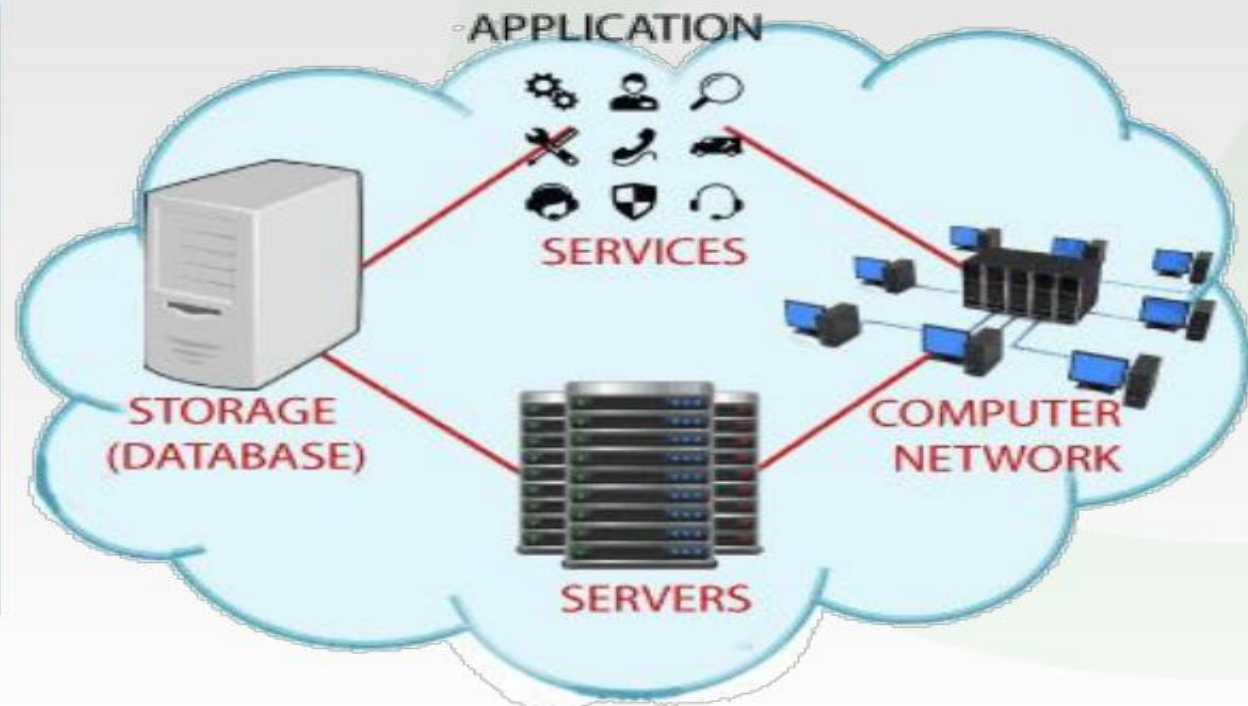
- Organizations of all types, sizes, and sectors use cloud computing for various purposes such as data backup, disaster recovery, email, virtual desktops, software development, testing, big data analytics, and customer-facing web applications.
- For example, healthcare companies use the cloud to develop more personalized treatments for patients. Financial service companies use the cloud to support real-time fraud detection and prevention. Video game developers use the cloud to provide online games to millions of players worldwide.

Introduction to Cloud Computing

Before
Cloud Computing



After
Cloud Computing

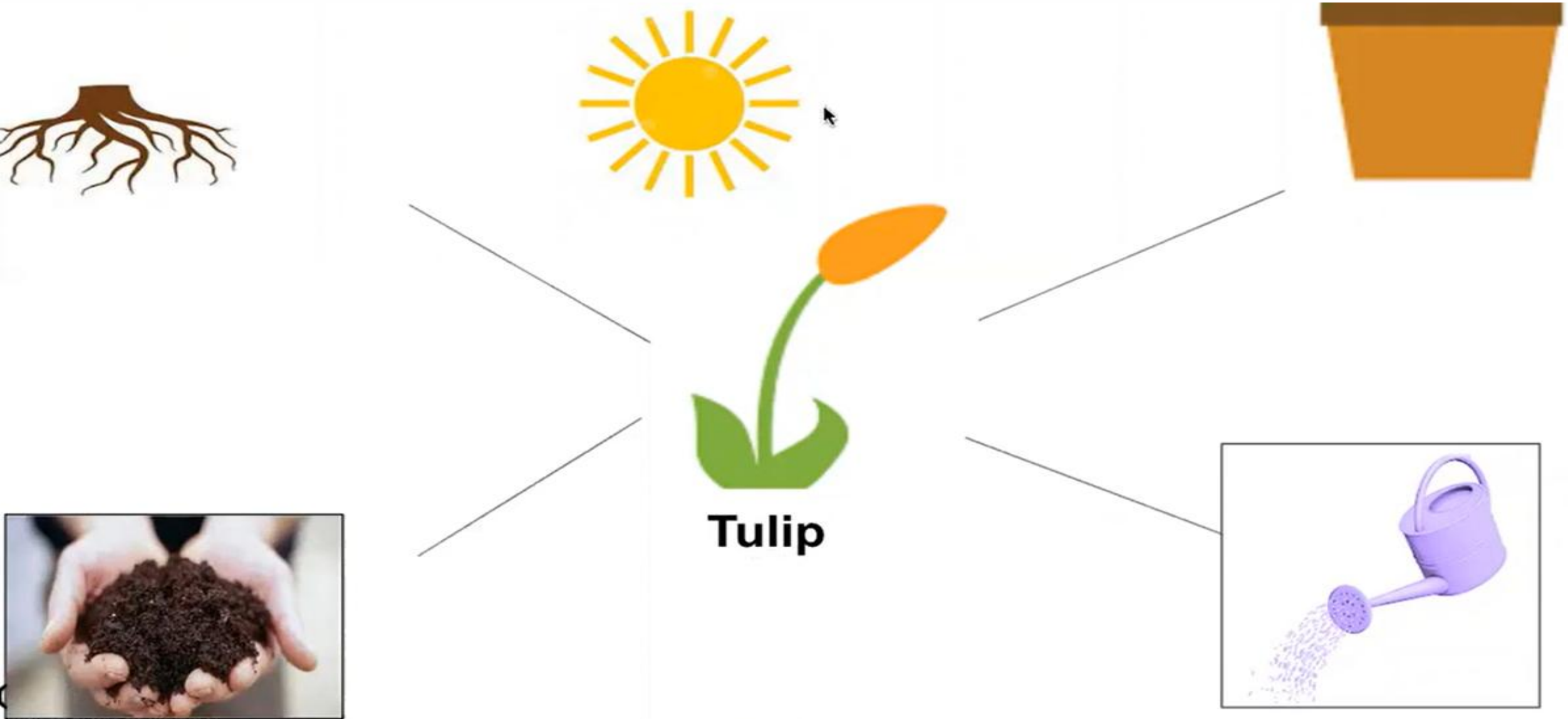




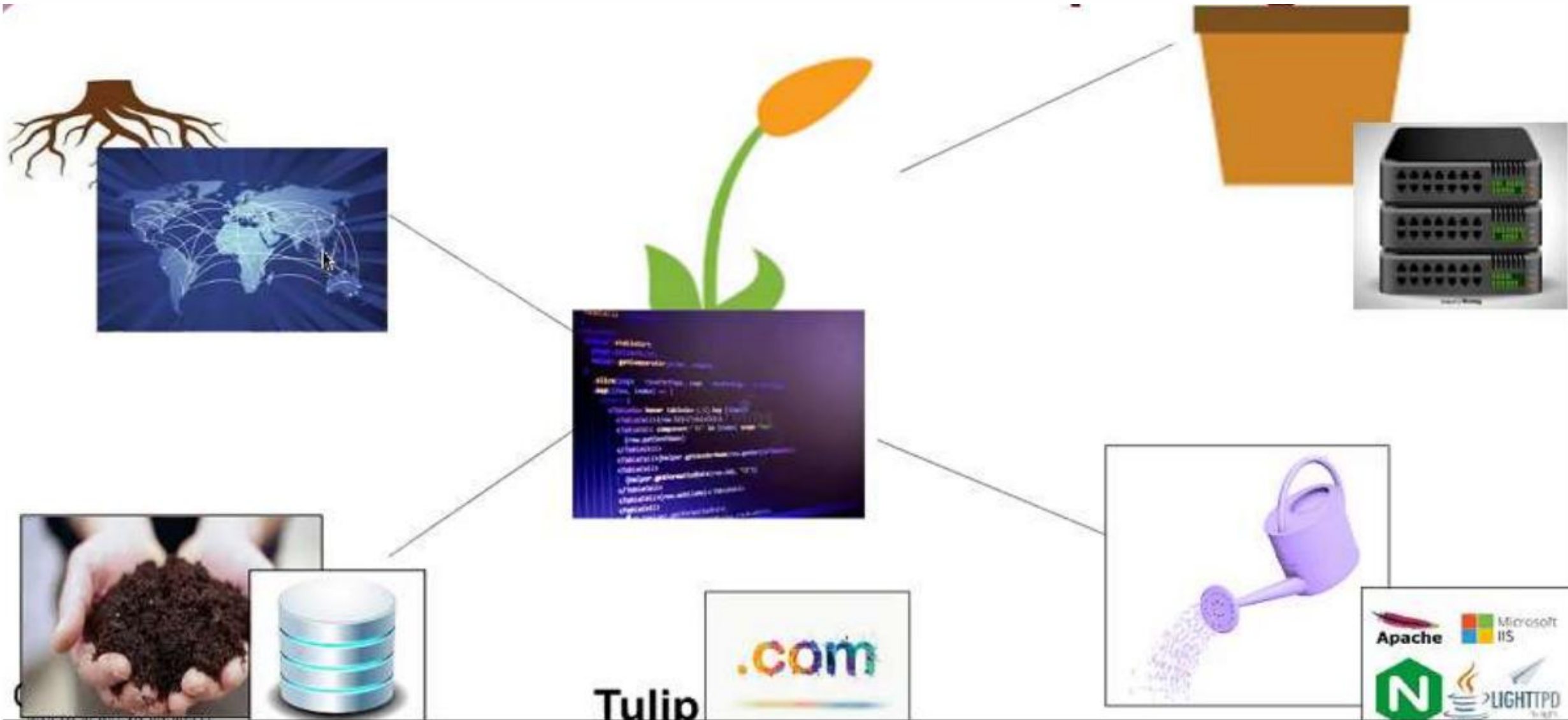
• **Why Cloud Computing?**

- Increases the value of the work
- Zeitgeist (The spirit of the time)
- Cost reduction (pay as you go - source optimization)
- Scalability need
- Virtualization
- Containerization Technology
- Software Development Cycle
- From Monolithic to Microservices

Introduction to Cloud Computing



Introduction to Cloud Computing



What is Cloud Computing?



Introduction to Cloud Computing

What is Cloud Computing?

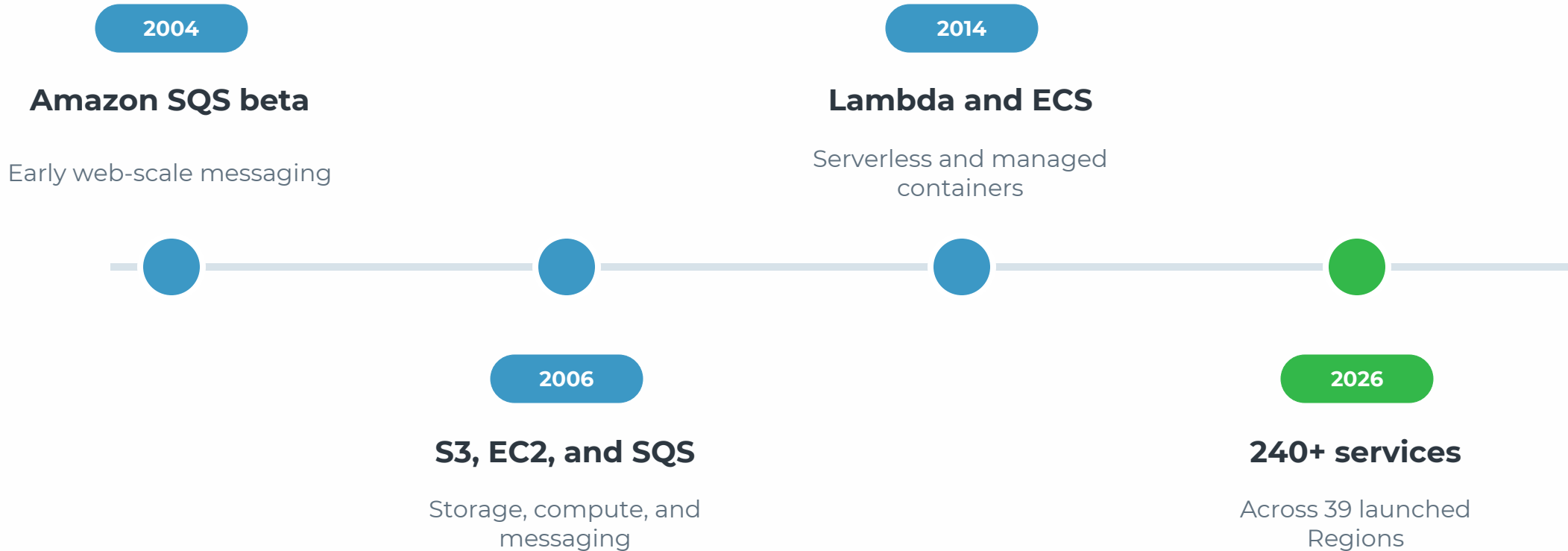
- The **Cloud** term refers to software and services running on the Internet, not locally on your computer.
- So you can store and access data and programs over the internet rather than the hard drive of your computer

Cloud Computing = Application running on someone else's computer



Cloud computing at AWS

Selected milestones that changed how teams build and operate systems



Introduction to Cloud Computing

What is Cloud Computing?

- In 2002, Amazon Web Services (AWS) launched its public cloud,



Key Features of Cloud Computing

- **High Availability and Flexibility:** You can access cloud resources from anywhere with the internet, using different devices. You can instantly scale the resources you need up or down.
- **Cost Efficiency:** Physical hardware investments and maintenance costs are reduced. You only pay for the resources you need, preventing resource waste.
- **Automation and Ease of Management:** Cloud services provide advanced tools to automate and manage workloads, simplifying IT management.
- **Security and Compliance:** Cloud providers offer advanced solutions for data security, encryption, network security, and compliance standards.
- **Disaster Recovery:** Data backup and recovery solutions ensure that data remains secure even in the event of a disaster.

Key Features of Cloud Computing



What is Cloud Computing?

Cloud Computing vs. Cloud Storage



Cloud
Storage



Cloud
Computing

Cloud

Advantages of the Cloud Computing

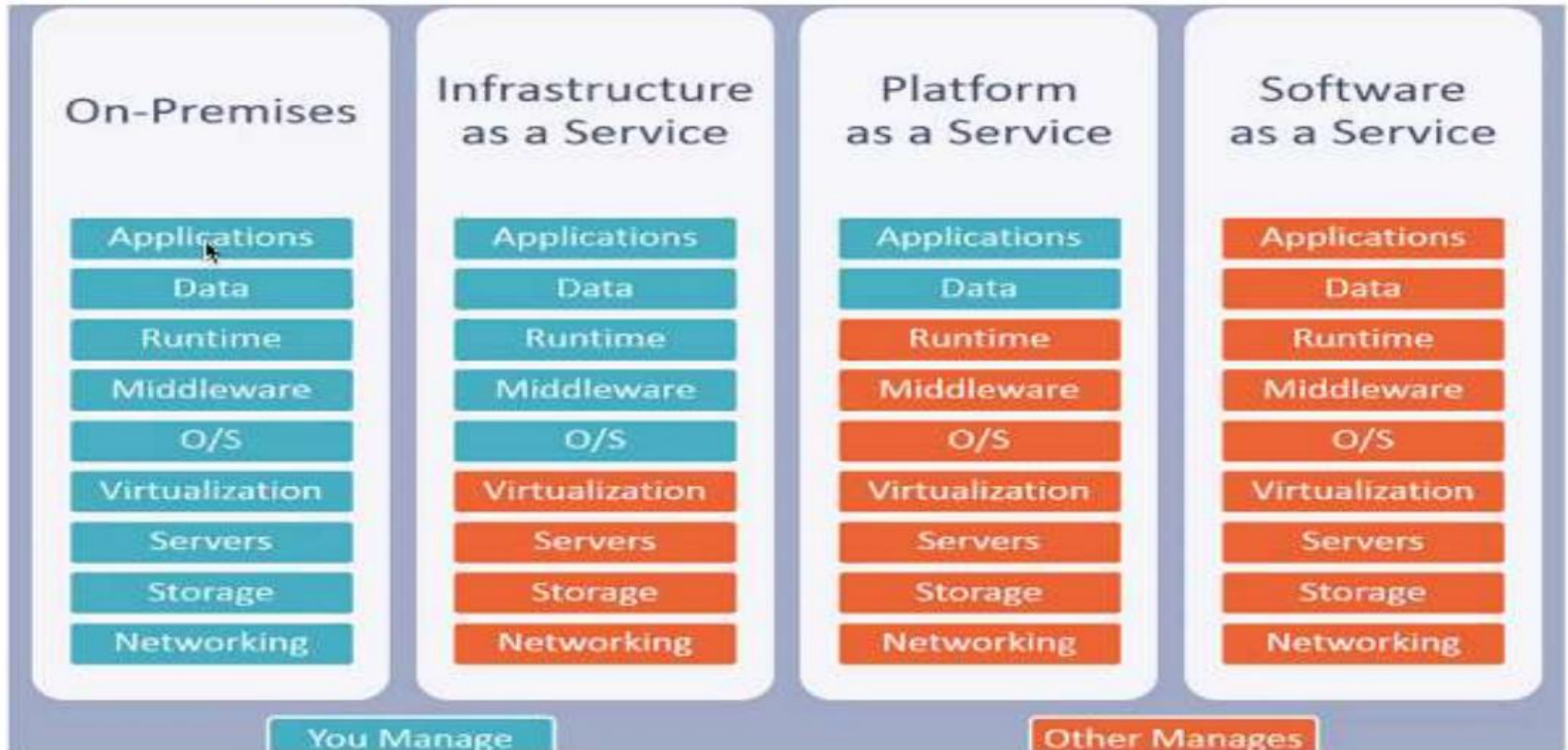
- Cost Efficiency
- Scalability
- Accessibility
- Disaster Recovery
- Security
- Collaboration
- Automatic Updates
- Environmental Impact
- Flexibility
- Innovation

Disadvantages of the Cloud Computing

- **Downtime**
- **Data Security and Privacy**
- **Limited Control and Flexibility**

Service Models

Cloud Service Models



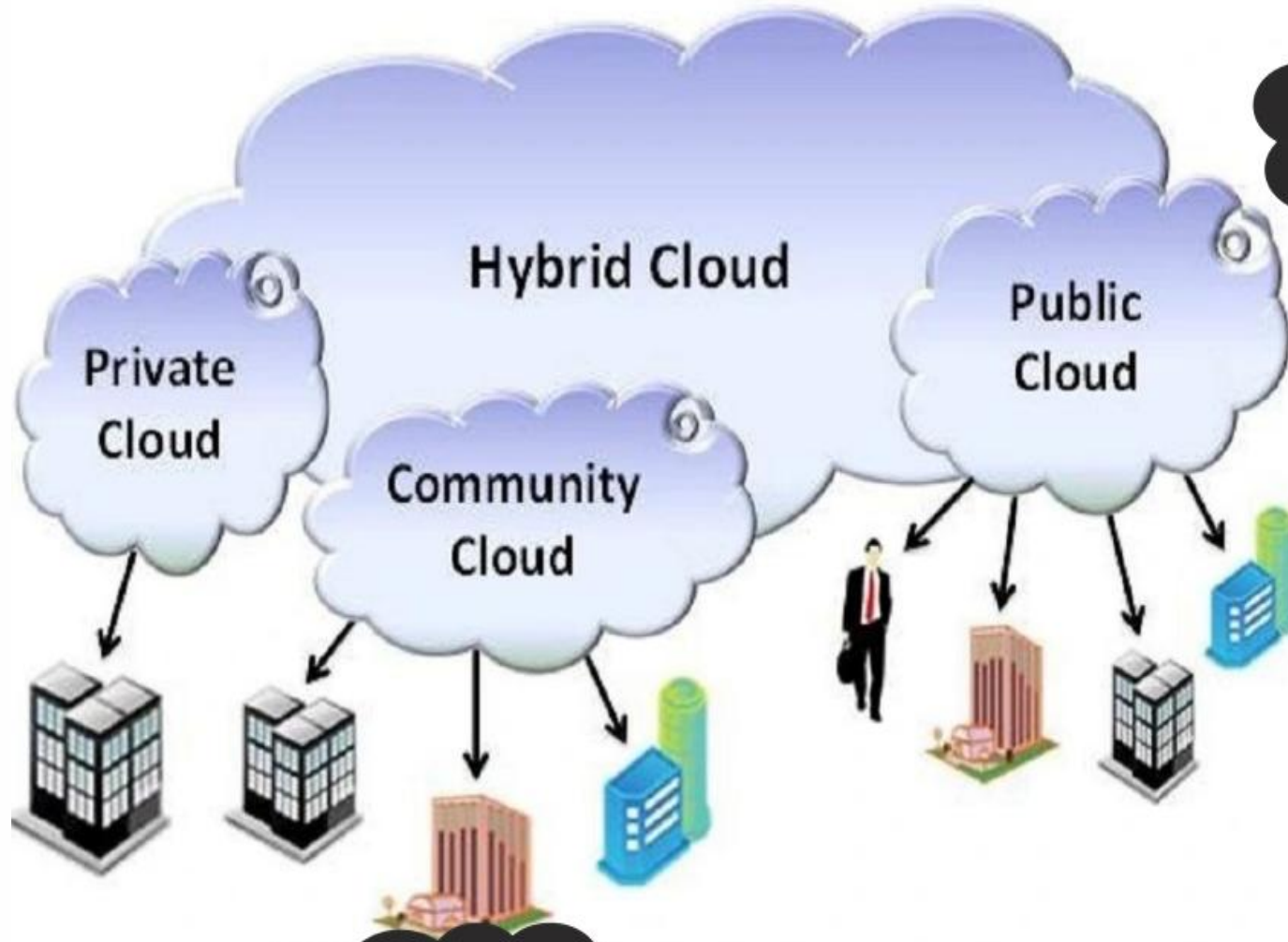
CAN YOU
DO HOMEWORK
AS A
SERVICE?



Deployment Models



IBM, Google,
Amazon,
Microsoft



E-Devlet
Financial
Institutions

Cloud Deployment Models

HYBRID

Combination of cloud deployment models

COMMUNITY

Manufacturing organization shares cloud with other organizations with similar interests

PUBLIC

Manufacturing organization shares cloud with general public

PRIVATE

Manufacturing organization has its own private cloud

Introduction to AWS

What is AWS?



- AWS stands for **A**ma**z**on **W**eb **S**ervices.
- Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted [Cloud Platform](#).

AWS at 20 years

Official scale indicators—not a third-party market-share snapshot

20

years since Amazon S3 and
Amazon EC2 launched

240+

cloud services in the AWS
portfolio

39

launched AWS Regions
worldwide

AWS describes its customer base as millions of customers worldwide.

Use the global infrastructure page for live counts; these figures change as AWS expands.

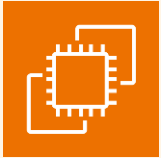
Some Features Provided by AWS:

- **Computing Power:** AWS provides the computing power needed for computational tasks over the internet. This allows heavy operations to be performed without overloading your computer.
- **Storage:** AWS offers a secure online storage space for your photos, videos, or important files.
- **Databases:** AWS provides the tools needed to store and organize data, ensuring that the data is kept in an organized and secure manner.
- **Security:** AWS offers many features to ensure the security of stored data.
- **Ease of Use:** AWS is customizable to meet user needs and is easy to use. You can increase or decrease resources whenever you want.



Core AWS building blocks

Learn the job each service performs before memorizing product names



Amazon EC2

Virtual servers



AWS Lambda

Run code on demand



Amazon S3

Object storage



Amazon RDS

Managed relational
DB



Amazon VPC

Isolated networking



AWS IAM

Identity and access

AWS service milestones—and name changes

Historical launch names are not always the names learners see today

2006

Amazon S3, Amazon EC2, and Amazon SQS

Storage • compute • messaging

2009

Amazon VPC and Amazon RDS

Networking • managed databases

2011

AWS CloudFormation

Infrastructure as code

2012

Amazon DynamoDB and Amazon Redshift

NoSQL • data warehousing

2014

AWS Lambda and Amazon ECS

Serverless • containers

Now

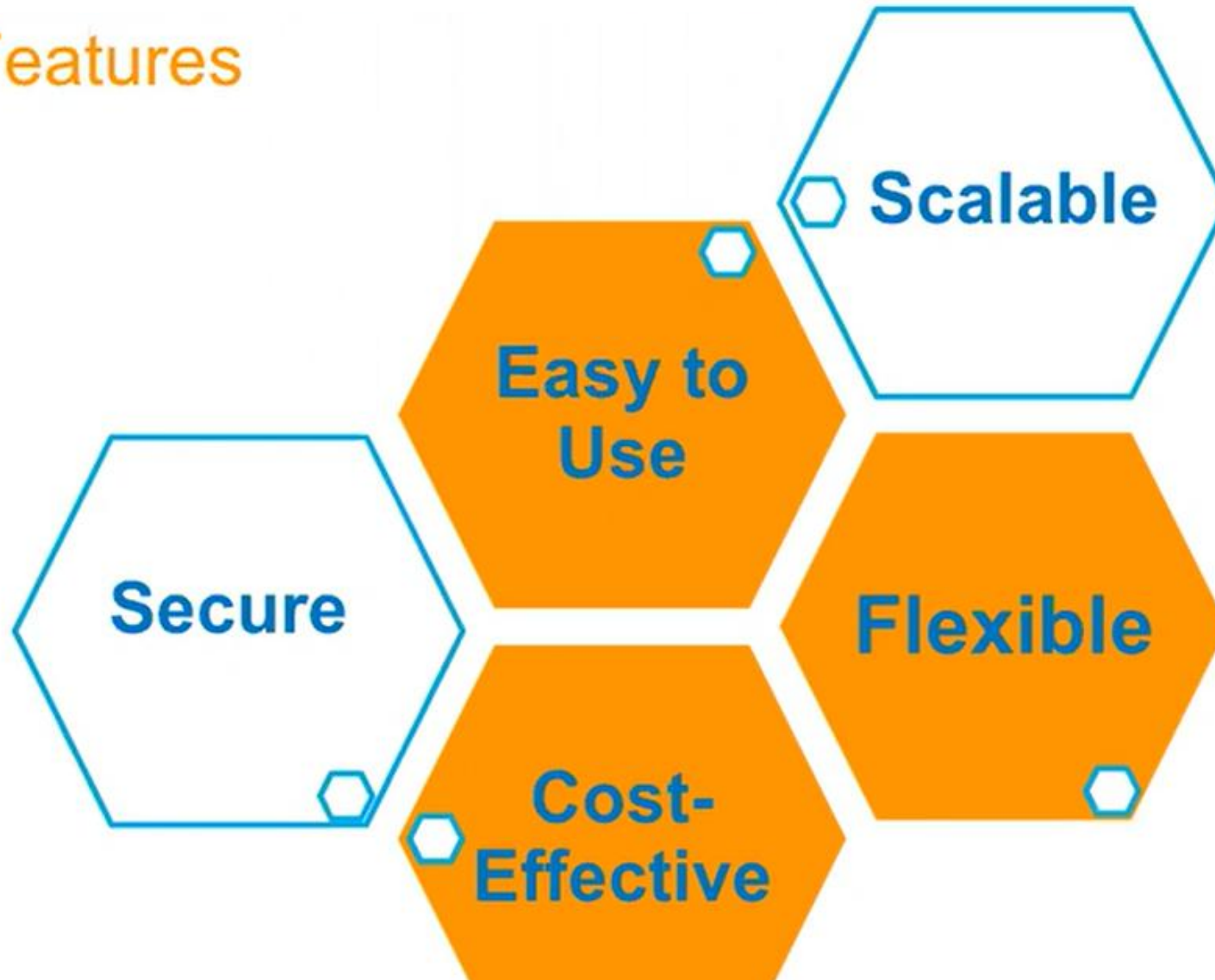
Use current names

Amazon ECS; S3 Glacier storage classes

The original Amazon Glacier vault service stopped accepting new customers on 15 Dec 2025.

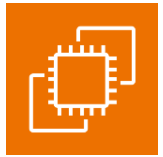
Introduction to AWS

AWS Features



How AWS services fit together

Six categories provide a durable mental model for the course



Compute

Run applications



Storage

Keep objects and files



Databases

Persist structured data



Networking

Connect and isolate



Security

Control identities and access



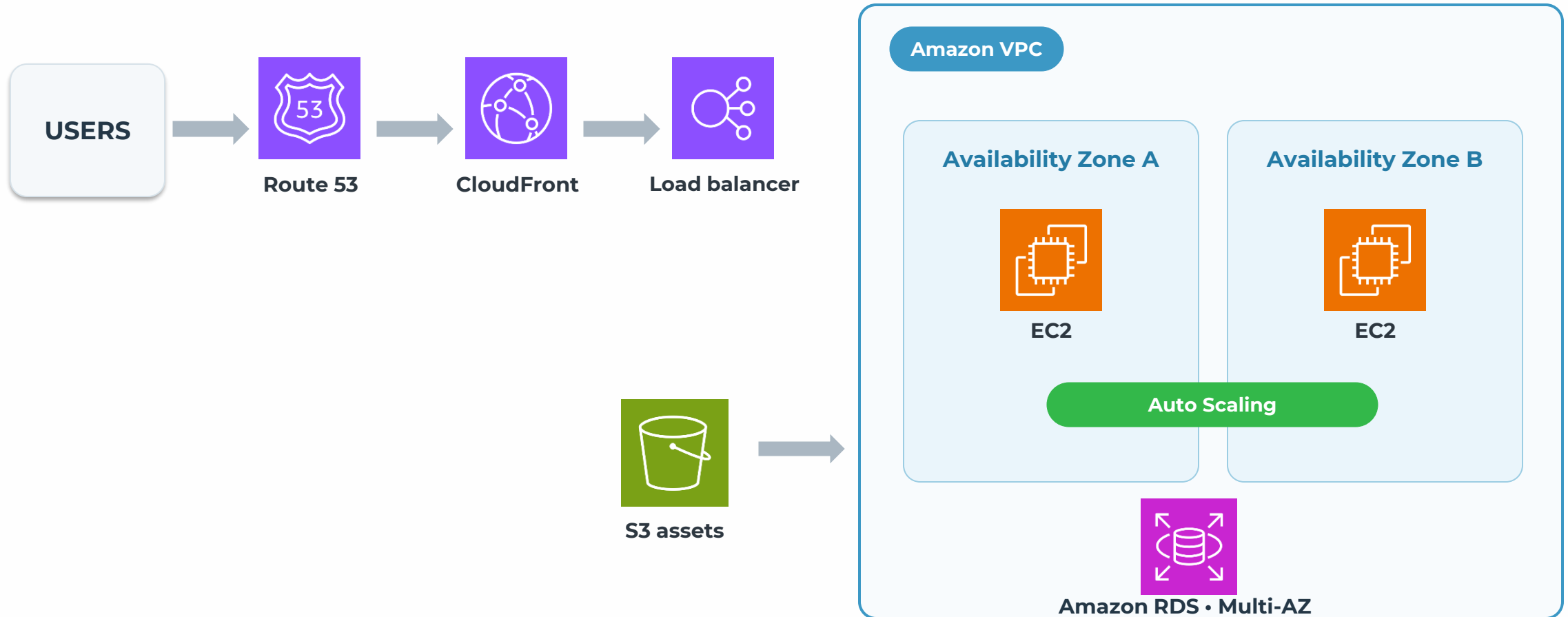
Management

Provision and operate

Architectures combine categories; a service rarely solves an entire workload by itself.

Reference architecture: highly available web application

A simple path from users to a multi-AZ workload

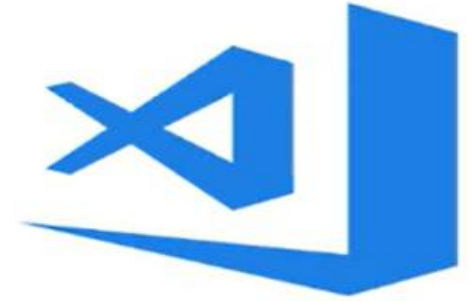


Route traffic • cache content • balance requests • scale across AZs • keep durable data

Problem,bug vs..



Developer



Change the code
Versioning CSS, HTML



Git Commit



Hook
function



Jenkins



Build &
Deploy
Pipeline
Send-EC2 server



Publish in Public
Internet



AWS Global Infrastructure

The live AWS infrastructure page is the source of record

39

launched Regions

124

Availability Zones

750+

CloudFront points of
presence

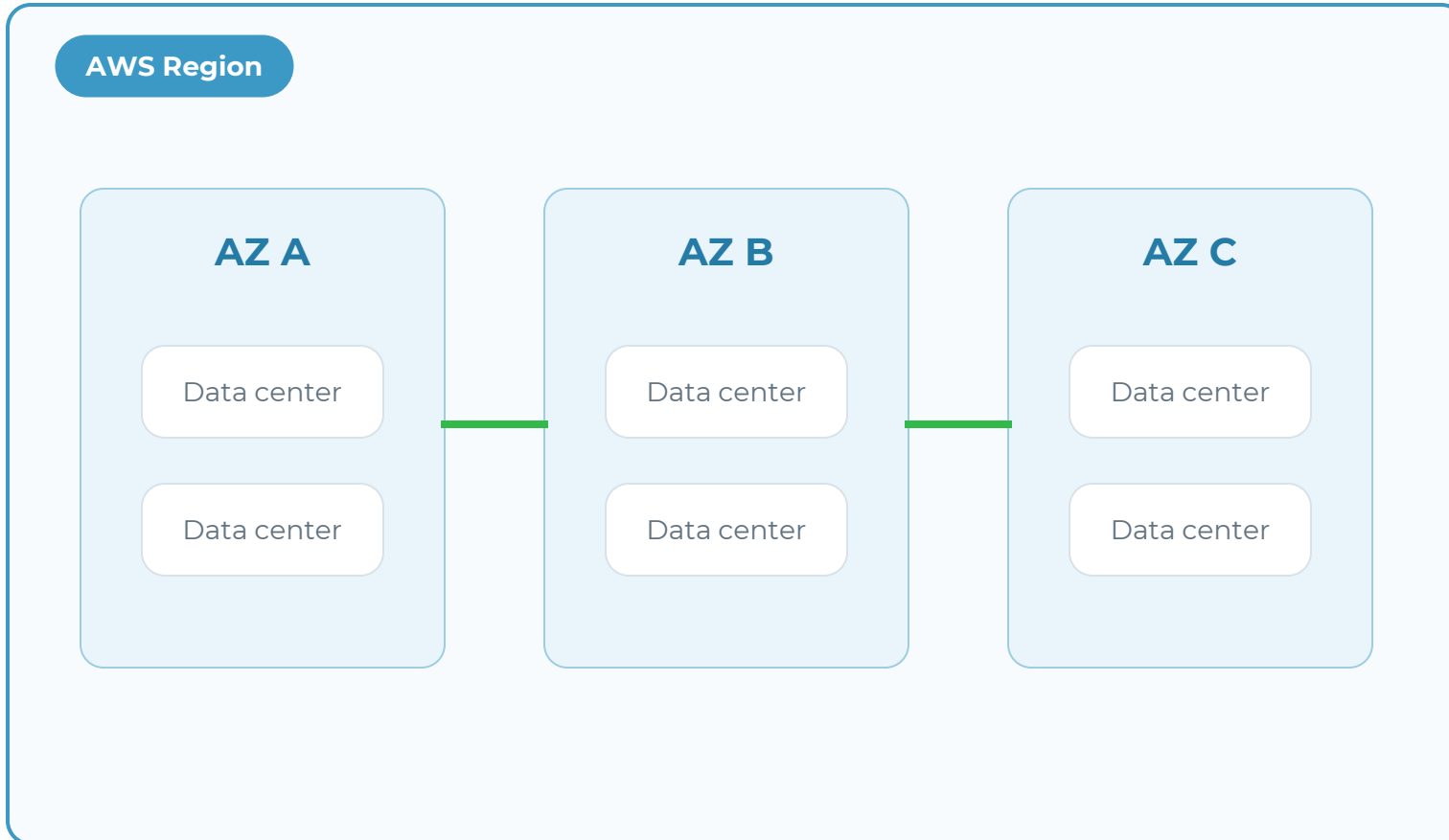
15

regional edge caches

Counts change as AWS launches infrastructure. Verify before teaching or publishing.

Availability Zones: independent by design

A Region contains multiple isolated locations connected by private high-speed links



Each AZ

- One or more discrete data centers
- Independent power, cooling, and networking
- Physically separated—up to about 100 km
- Connected with low-latency private links

Design across ≥ 2 AZs

Regions and Availability Zones

Choose a Region deliberately; design resilient workloads across Availability Zones

1 Choose a Region

Evaluate:

- Data residency and compliance
- Latency to users
- Service and feature availability
- Pricing and sustainability needs



2 Use multiple AZs

For high availability:

- Place redundant resources in ≥ 2 AZs
- Route around component failure
- Replicate data appropriately
- Test recovery, not just deployment

A Region choice is an architecture decision; an AZ choice is a resilience decision.

AWS Infrastructure

AWS Data Centers

The data center infrastructure consists of **physical or hardware-based tools and components**.

AWS data centers are **secure** by design.



Northern Virginia Data Center

AWS Infrastructure

AWS Global Infrastructure



AWS edge network

Bring content, DNS answers, and application entry points closer to users

750+

CloudFront points of presence

15

regional edge caches

Global

network connecting users to AWS



Route 53

DNS routing



CloudFront

Cached delivery



AWS workload

Origin services

AWS Shared Responsibility Model

Responsibility shifts with the service—but it never disappears

AWS • security OF the cloud

Facilities and physical security
Hardware, networking, and the global infrastructure
Managed service infrastructure and foundational controls

Customer • security IN the cloud

Data classification and protection • identities and permissions • network configuration
Guest OS and application patching where applicable • encryption choices • secure code

The exact split depends on the service model: IaaS, containers, managed services, or serverless.

AWS Certification pathways

Choose a path by role and experience—not by collecting every badge

FOUNDATIONAL

Cloud Practitioner

AI Practitioner

ASSOCIATE

Solutions Architect

Developer

CloudOps Engineer

Data Engineer

Machine Learning
Engineer

PROFESSIONAL

Solutions Architect

DevOps Engineer

Generative AI Developer

SPECIALTY

Security

Advanced Networking • retires 31 Dec 2026

Operations certification renamed: AWS Certified CloudOps Engineer – Associate.

AWS Free Tier for new accounts

For accounts created on or after 15 July 2025

Both plans include \$100 at signup + opportunities to earn up to \$100 more in credits, plus 30+ services with monthly free usage limits.

FREE PLAN • BOOTCAMP START

- No charges while the plan is active
- Access to selected services and features
- Ends after 6 months or when credits are used
- Account closes automatically when the plan ends
- Upgrade later when a lab requires paid access

PAID PLAN

- Access to all AWS services and features
- Pay-as-you-go beyond credits or ineligible usage
- Account remains open after credits are depleted
- Eligible for short-term trials and more promotions
- Better fit for production and unrestricted labs

Plan choice affects access, cost behavior, and later setup decisions.



HANDS-ON PREPARATION

Creating an AWS Account

Advanced signup • Free plan first • Secure before labs

Console labels may evolve; follow the task names and verification checks in the lab guide.

1. Choose the signup path

AWS may show different signup experiences during a limited rollout

USE FOR THIS COURSE

Advanced signup

- Full account control
- Root user and IAM access
- All services allowed by the selected plan
- Best match for bootcamp labs

LIMITED ROLLOUT

New signup experience

- Simplified onboarding
- May restrict services or account capabilities
- Not appropriate when a lab needs full control

If AWS does not offer a choice, follow the path presented and confirm the readiness checklist on slide 52.

2. Create the root user

Use an email address and recovery controls you can keep secure

- 1 Root email** Use a monitored address that is not shared casually.
- 2 Account name** Choose a recognizable training-account name.
- 3 Verify email** Enter the code AWS sends to the root email.
- 4 Strong password** Use a unique password stored in a password manager.

Root credentials are for account-level tasks—not everyday console work.

3. Select the Free plan

Start with guardrails; upgrade only when a later lab needs unrestricted access

BOOTCAMP RECOMMENDATION

Free plan

**\$100 signup credit
+ up to \$100 earned
credits**

No charges while active
Selected services and features
Ends after 6 months or credit
depletion

Do not enable AWS Organizations or Control Tower on the Free plan: AWS says this automatically upgrades the account and can expire credits immediately.

4. Enter contact and payment details

The exact fields, tax identifiers, and payment methods vary by country

1 Account type

Personal or business—choose the option that matches the account owner.

2 Contact details

Enter a reachable address and phone number for verification and recovery.

3 Payment method

Provide a supported card or payment method for identity verification.

4 Tax information

Complete only the fields AWS displays for your billing country.

A payment method is required even for the Free plan, but the Free plan itself prevents charges while active.

5. Verify identity and activate the account

Complete verification, choose support, and wait for the activation email

1 Phone verification

Choose SMS or voice when offered and enter the verification code.

2 Support plan

Select Basic Support for the training account.

3 Activation

Wait for AWS to confirm the account is ready; activation can take time.

4 First sign-in

Sign in as root only to perform the security setup on the next slide.

Current AWS Support plans: Basic, Business Support+, Enterprise Support, and Unified Operations.

6. Secure the root user immediately

Root-user MFA is required within 35 days of the first console sign-in



- 1 Open the account menu → Security credentials
- 2 Choose Assign MFA device for the root user
- 3 Prefer a passkey or hardware security key
- 4 Register more than one MFA device when possible
- 5 **Store recovery information securely; create no root access keys**

After this step, sign out of root. Use root again only for tasks that explicitly require it.

7. Create daily training access

Use temporary credentials in production; keep the Free-plan classroom exception explicit

PRODUCTION RECOMMENDATION



IAM Identity Center or federation

Human users receive temporary credentials through roles. This is the AWS-recommended pattern.

FREE-PLAN CLASSROOM FALLBACK



Console-only IAM administrator

Create one IAM user for the learner, attach AdministratorAccess through a group, require MFA, and create no access key.

Why the exception? Enabling an Organizations-backed Identity Center instance upgrades the Free plan and can expire credits.

8. Add billing safeguards

Monitoring is still required—even when the Free plan prevents charges



Free Tier dashboard

Monitor plan days remaining, credit balance, and service usage.



Zero spend budget

Billing and Cost Management → Budgets → Create budget → Use a template → Zero spend budget.



Email alerts

Use an email address the learner checks and confirm any required subscriptions.

A budget is an alerting mechanism—not a hard spending cap. Always clean up lab resources.

Account readiness checklist

Complete every item before starting resource-creation labs



Account activation email received



Free plan selected and expiry understood



Root user has MFA and a backup recovery path



Root access keys do not exist



Daily training user has console access and MFA



No access key created for the training user

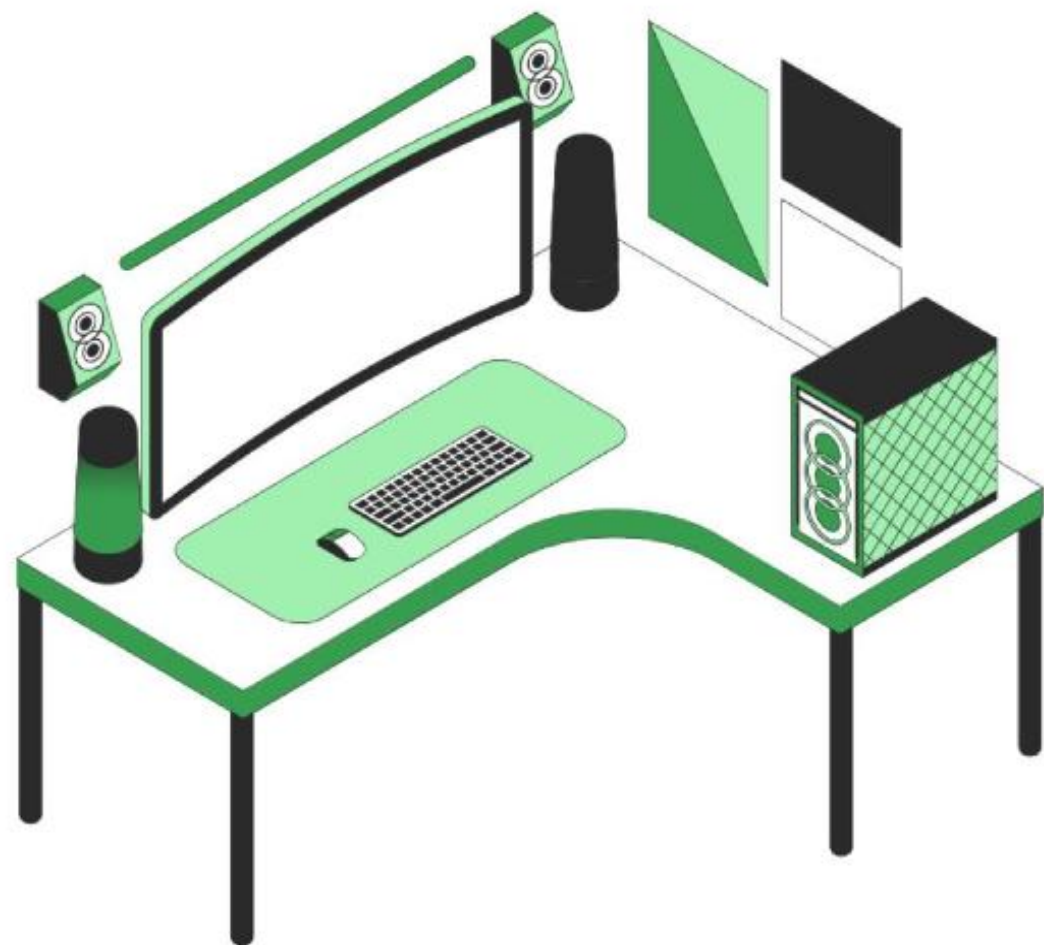


Zero spend budget and email recipient configured



Basic Support selected and a working Region chosen

READY FOR MODULE 2



Do you have any questions?

Send it to us! We hope you learned something new.